



Biological impacts of glyphosate on morphology, embryo biomechanics and larval behavior in zebrafish (*Danio rerio*)

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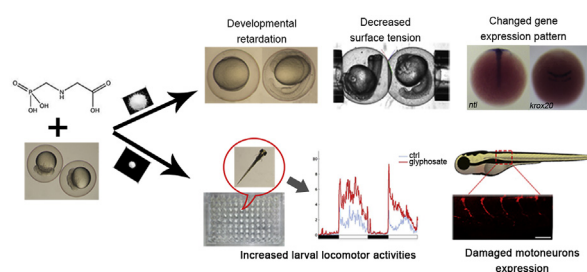
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HIGHLIGHTS

- High concentrations of glyphosate can induce zebrafish development delay and embryonic death.
- Biomechanics analysis indicates the surface tension of chorion is decreased after glyphosate exposure.
- Embryonic exposure to glyphosate damages embryos CaP axons and induces larval locomotor activities increased.

GRAPHICAL ABSTRACT



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ABSTRACT

All of these days, residues of herbicides such as glyphosate are widely distributed in the environment. The ubiquitous use of glyphosate has drawn extensive attention to its toxicity as an organic pollutant. In this study, we employed larval zebrafish as an animal model to evaluate the effect of different concentrations of glyphosate on early development via morphological, biomechanics, behavioral and physiological analyses. Morphological results showed that an obvious delay occurred in the epiboly process and body length, eye and head area were reduced at concentrations higher than 10 mg/L. The expression of *ntl* (*no tail*) shortened and *krox20* (*also known as Egr2b, early growth response 2b*) changed as the glyphosate concentration increased, but there was no change in the expression of *shh* (*sonic hedgehog*). In addition, biomechanical analysis of the elasticity of chorion indicated that treated embryos' surface tension was declined. Furthermore, a 48-h locomotion test revealed that embryonic exposure to glyphosate significantly elevated locomotor activities, which is probably attributed to motoneuronal damage. The decreased surface tension of chorion and the increased locomotive activities may contribute to the hatching rates after glyphosate treatment. Our study enriches the researches of evaluating glyphosate toxicity and probably plays a warning role in herbicides used in farming.

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1. Introduction

As the active ingredient of more than 750 different broad-

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spectrum herbicides, glyphosate is widely used in agriculture because of its effective control of weeds (Mesnage et al., 2015). The glyphosate-based herbicide (common trade name “Roundup”) was first sold to farmers in 1974 (Myers et al., 2016). Since the late 1970s, the volume of glyphosate-based herbicides (GBHs) applied has increased approximately 100-fold (Myers et al., 2016). According to recent findings, more than 600,000 tons of glyphosate have been used globally (Wang et al., 2016). Considering its short half-life and microbial degradation, it will be rapid inactivated in soil (Busse et al., 2001; Mesnage et al., 2015). So glyphosate has been usually thought to be environmentally safe. At sub-agricultural concentrations, glyphosate has little effects on organisms (Bringolf et al., 2007; Gasnier and Dumont, 2009; Akcha et al., 2012; Fan et al., 2013; Jin et al., 2013). Although its mechanism of removing competing vegetation targets an enzyme found only in plants and certain bacteria (Dill, 2005), The toxicity of glyphosate to vertebrates is not fully understood. Previous research has substantially discovered toxicological impacts of glyphosate. Glyphosate-based herbicides produce teratogenic effects on vertebrates -*Xenopus laevis* and chicken (White Leghorn strain) embryos, by impairing retinoic acid signaling (Alejandra Paganelli et al., 2010). Roundup® exposure can increase superoxide dismutase (SOD), glutathione peroxidase (GPx) activity, lipid peroxidation (LPO) and reduce catalase (CAT) activity on Asian clam *Corbicula fluminea* (dos Santos and Martinez, 2014). Glyphosate induced toxic effects in nontarget organisms such as brine shrimp (*A. salina*) nauplii (de Brito Rodrigues et al., 2016) and zebrafish early life stages (Roy et al., 2016b). In addition, breeding zebrafish exposed to 10 mg/L glyphosate during a period of 21 days reduced egg production except fertilization rate, and increased early stage embryo mortalities and premature hatching. Transcript profiling of the gonads revealed 10 mg/L Roundup and glyphosate induced changes in the expression of *cyp19a1* and *esr1* in the ovary and *hsd3b2*, *cat*, and *sod1* in the testis (Uren Webster et al., 2014). One study demonstrated that glyphosate product (Roundup®) might lead to excessive extracellular glutamate levels and consequently to glutamate excitotoxicity and oxidative stress, caused neurotoxicity in hippocampus of immature Wistar rats (Cattani et al., 2014). All in all, several studies have suggested that glyphosate may act as a teratogen, an endocrine disruptor and as a carcinogen (Pandey and Rudraiah, 2015; Claudinei et al., 2016; Myers et al., 2016). In 2005, the Food and Agriculture Organisation (FAO) reported that glyphosate and its major metabolite, aminomethylphosphonic acid (AMPA), are of potential toxicological concern, mainly as a result of accumulation of residues in the food chain (Bai and Ogbourne, 2016). So, this issue should arouse wide concern and deserve our attention.

Zebrafish have several inherent advantages for experiments. They are small, inexpensive to maintain and easily bred in large numbers – a single spawning produces 100–200 eggs. In addition, administration of drugs is simple. Zebrafish larvae absorb small molecules diluted in the surrounding water through their skin and gills. As a predictive animal model, zebrafish has been used in a variety of assays for assessing cardiotoxicity, hepatotoxicity, neurotoxicity and developmental toxicity (McGrath and Li, 2008; de Esch et al., 2012). Researchers used the rest/wake behaviour of larval zebrafish to study psychoactive drugs (Rihel et al., 2010). Someone assessed the toxicity of compounds according to the behavioral profiling (Li et al., 2014; Wang et al., 2015). Previous research on the mechanical properties of the chorion focused on the change in shape and elastic modulus of the zebrafish chorion under perturbation by an external force. The chorion is described as viscoelastic sphere (Kim et al., 2004; Lu et al., 2009; Nam et al., 2010). Here, we focused on the surface tension of chorion to detect the damage of chorion.

It is reported that annual farm-sector glyphosate usage reached to approximately 240 million pounds (~108.8 million kilograms) by 2014 (Myers et al., 2016). Worldwide, glyphosate concentrations can reach up to ~10–15 µg/L averagely in more diluting bodies of water, like rivers (Uren Webster et al., 2014; Roy et al., 2016a). The maximum concentration of glyphosate in surface water reached to 40.8 µg/L in the Orge watershed (France). Maximum concentrations (75–90 µg/L) were detected during application period and rainfall event in storm sewer and in wastewater sewer due to overflows (Botta et al., 2009). Levels of glyphosate in waters ranged from 0.10 to 0.70 mg/L, while in sediments and soils values were between 0.5 and 5.0 mg/Kg in the field from a transgenic soybean cultivation area located in the north of the Province of Buenos Aires, Argentina (Peruzzo et al., 2008). Glyphosate is easy to contaminate the surface runoff, groundwater and farmland because of its high solubility in water (12 g/L at 25 °C) and high sorption (>1.000 L/kg) to soil particles (Jin et al., 2013). Although the concentrations in the normal environment are very low, the concentrations of glyphosate from farmland orchards and ponds are possibly higher due to the glyphosate application and lack of water flow. With regard to the highest concentration, it mainly associated with direct aquatic application and in isolated environments (Uren Webster et al., 2014). Furthermore, if farmers and children were exposed to glyphosate herbicide directly, they would suffer serious damages. Previous studies proved that environmentally relevant concentrations of glyphosate have induced different damages to organisms in some respects. The treatment concentrations in our study included both the environmental relevant concentrations and impossible concentrations in extreme cases. Here, we added higher concentrations of glyphosate to explore its toxicity. Zebrafish was chosen as the laboratory animal model to access the toxicity of glyphosate on organisms under the comprehensive concentrations. Initially, developmental, morphological and genetic effects of glyphosate on zebrafish embryos from 3 hpf (hours post fertilization) to 96 hpf were evaluated. Subsequently, biomechanical impacts induced by glyphosate were examined by quantifying the surface tension of embryos exposed to different concentrations of glyphosate at 3, 10 and 24 hpf. Furthermore, the automatic, high-throughput video-tracking system was introduced to observe the behavioral effect caused by glyphosate via continuously monitoring the locomotion of zebrafish over 48 h. Finally, locomotion increase was probably attributed to the alteration of larval motoneuron at 28 hpf by immunofluorescence assay.

2. Materials and methods

2.1. Zebrafish maintenance

The zebrafish (AB strain) were raised at a constant temperature of 28.5 °C on a constant light cycle (14 h light/10 h dark). The fish-farming system was equipped with circulating water to maintain a standard water system (KCl 0.05 g/L, NaHCO₃ 0.025 g/L, NaCl 3.5 g/L, and CaCl₂ 0.1 g/L, with 1 µg/mL methylene blue, pH 7.0–7.3). The zebrafish were fed fresh brine shrimp twice every day. In the night before experiment, one male and one female zebrafish were transferred to a breeding tank with a sliding door. The next morning, the sliding door was removed to allow the male to pursue the female. Sufficient eggs were collected in about 15 min after fertilization. The eggs were washed to remove the unfertilized eggs and debris. The normal fertilized eggs were used for subsequent experiments. All experimental protocols and procedures involving zebrafish were approved by the Committee for Animal Experimentation of the College of Life Sciences at Nankai University (no. 2008) and were performed in accordance with the NIH Guide for the Care and Use of Laboratory Animals (no. 8023, revised in 1996).

2.2. Experiment design and chemical exposure

We conducted a series of experiments to detect biological effects of glyphosate on zebrafish (Supplementary Fig. 1A). Glyphosate [N-(phosphonomethyl)glycine] (Duke and Powles, 2008), as a weak acid, is easily dissolved in water. To simulate the environment, we dissolved glyphosate powder in the standard system water. To determine the diameter distribution and zeta-potential of the treatment concentrations in standard system water, we conducted the dynamic light scattering (DLS). Besides, we tested the pH of treatment concentrations with pen pH meter PHB-3(Shanghai) before glyphosate exposure.

2.3. Glyphosate exposures in zebrafish embryonic development

The glyphosate was purchased from Beijing Qinchengyixun Technology Co. Ltd (Beijing, China). The purity of Glyphosate was to 99.8%. We weighed and added powdered glyphosate directly to the standard system water to prepare stock solutions of 600 mg/L. Then we diluted it to obtain different concentrations (600, 400, 200, 100, 10, 5, 1, 0.5, 0.1, 0.01 mg/L) according to different experiments. All glyphosate solutions were obtained freshly just prior to the experiments. The collected embryos were transferred to 6-well plate, 40 embryos in each well. The standard system water was removed from each well, and then appropriate dilution was added immediately. Glyphosate exposure was conducted at 0.75 hpf until 96 hpf and solutions were refreshed daily. During this period, embryo mortality was recorded at 3, 6, 10, 24, 48, 72 and 96 hpf. The percentage of epiboly was counted at 10 hpf and embryo hatching rate was recorded at 48 and 72 hpf. All embryos were observed under a stereomicroscope (Olympus ZX-10, Japan) equipped with a digital camera (Canon) at 3, 6, 10, 24 hpf and photographed for morphological analyses. At 3, 10, 24 hpf, choosing three different concentrations (1, 10, 100 mg/L) to test the surface tension of embryo. At 10 and 13 hpf, four treatment groups (0.1, 1, 10, 100 mg/L) were fixed with 4% paraformaldehyde (PFA) for whole-mount *in situ* hybridization. At 48 hpf, six treatment groups (0.01, 0.1, 0.5, 1, 5, 10 mg/L) were rinsed and the cultured in clean standard system water until 96 hpf for diel activity behavior assay. All experiments were conducted three times and had a control group (0 mg/L).

2.4. Morphological analyses of the larvae

Glyphosate exposure was conducted until 96 hpf and solutions were refreshed daily. At 96 hpf, all survived larvae were anesthetized by immersed in tricaine for photography under a stereomicroscope (Olympus ZX-10, Japan) equipped with a digital camera (Canon). Larvae were immobilized in low density agar gels to ensure photographs were taken from a consistent orientation. The body length and the head and eye areas of the larvae were measured using ImageJ software.

2.5. The test of embryos' surface tension

We chose three different concentrations (1, 10, 100 mg/L) to test the ratio of tension at a normal and a treated embryos' chorion. And each concentration is divided into three groups whose treatment periods are 3, 10, 24 hpf, respectively. A normal embryo and a treated embryo are held by two capillaries of micromanipulator with similar and low pressure. The two embryos were moved to the position where they slightly contact and their symmetry axis were located on one line which was in the same horizontal plane and paralleled with micromanipulator movement trajectory. Then the right arm of micromanipulator moved 200 μm forward to induce shape change of chorion. The micrographs were took after 5 min of

squeezing. The angles θ_l and θ_r between the tangent of chorion-chorion interface and the tangent of left or right chorion-water interface at the edge of interface were determined by using edge detection program.

2.6. Whole-mount *in situ* hybridization

Embryos of different groups (0, 0.1, 1, 10, 100 mg/L) were collected respectively at 10 hpf (for *ntl*) and 13 hpf (for *krox20* and *shh*) for whole mount *in situ* hybridization and fixed in 4% paraformaldehyde, pH 7.0, in phosphate-buffered saline solution overnight at 4 °C. 20 embryos were collected for test at every group. Each experiment was repeated three times. Plasmids containing *ntl*, *krox20* and *shh* were used for probes synthesis. Antisense probes and whole-mount *in situ* hybridization were performed as previously described (Li et al., 2016a).

2.7. Diel activity behavior assay of larvae

First we decided to use six lower concentrations (0.01, 0.1, 0.5, 1, 5, 10 mg/L) to test the effects of glyphosate on zebrafish larval locomotor activity behavior. We established a video-tracking system to record their locomotor activity over 48 continuous hours. The larvae were pipetted into a 96-well plate, one larva per well. Each group tested twelve larvae. To minimise evaporation, the 96-well plate was sealed with a layer of plastic film and then put on an acrylic shelf. An infrared LED array was located below the shelf as a backlight source. An acrylic diffuser was placed above the infrared LED array. A camera (MV-VS078FM, MicroVision, Japan) with a fixed-angle megapixel lens (MP5018, Computer) was fixed on the top for vertical observations. The camera can yield high-quality images with a resolution of 1024*768 at 15 fps. A custom program was developed with Microsoft Visual Studio 2010 and OpenCV 2.4.3 to record the larval diel activity behavior. The algorithm was based on the subtraction of the adjacent frames.

During the experiment, the whole system was in a room at a constant temperature of 28.5 °C and with illumination from 7:00 a.m. to 9:00 p.m. We began the test at 4dpf (days post fertilization) at 9:00 p.m.

2.8. Immunofluorescence

Whole-mount immunofluorescence was performed using standard zebrafish protocols (Li et al., 2016b). Larvae (28 hpf) of different treated groups (0.01–10 mg/L) were collected and fixed overnight at 4 °C in 4% paraformaldehyde, pH 7.0, in phosphate-buffered saline solution. $n = 7$ larvae. After washing by PBST, the embryos were kept in cold acetone. After incubation in 10% sheep serum for 1 h, the larvae were immersed in a primary antibody Znp1 (1:400) antibody overnight at 4 °C. Following several washes with PBST, larvae were incubated with a secondary antibody Cy3-labeled goat anti-mouse IgG diluted 1:300. Finally Laser Confocal Scanning Microscopy was used to observe the expression of fluorescence.

2.9. Data analyses

The composition of surface tension will be zero when solid or liquid attains equilibrium (i.e. minimum surface energy), such as liquid will maintain perfect spheres at zero gravity circumstances. Here, the chorion is described as isotropic elastic sphere. When two embryos contact each other and their shapes change, three kinds of surface tensions (tension at left chorion-water interface T_l , tension at right chorion-water interface T_r , tension at chorion-chorion interface T_i) exist on the edge of interface of two embryos. The

composition of three surface tension will be zero when the shape of chorion doesn't change and their interface doesn't move. Because of this relation between the three surface tension, we use three vectors which can form a triangle to substitute the three tension and the sum of three vectors is 0.

$$\vec{T}_r + \vec{T}_l + \vec{T}_i = 0 \quad (1)$$

So we can get:

$$\frac{T_r}{\sin \theta_l} = \frac{T_l}{\sin \theta_r} = \frac{T_i}{\sin \theta_i} \quad (2)$$

$$\frac{T_r}{T_l} = \frac{\sin \theta_l}{\sin \theta_r} \quad (3)$$

The angles θ_l and θ_r between the tangent of chorion-chorion interface and the tangent of left or right chorion-water interface at the edge of interface were determined by using edge detection program. So we can calculate the ratio of tension at normal and treated embryo's chorion with Eq. (3).

From the diel activity behavior assay over 48 h, we obtained a time series from the raw data of the offline file. Each number in the time series represented the number of movements by one larva in 1 min during the 48 consecutive hours. To assess the larval locomotor activity changes, five parameters were extracted from the raw data: the number of rest bouts, rest bouts length, rest total, total activity, and waking activity. These parameters were

calculated for each experimental day and night. Any minute with less than 0.3 s of total movement was defined as one rest minute. A rest bout was defined as a continuous series of rest minutes. The rest total was defined as the total minutes of rest during all experimental days and nights. The rest bout length was defined as the average length of a continuous string of rest minutes. The total activity was defined as the average amount of activity per animal. Waking activity was also defined as the average amount of activity after excluding all rest minutes (Wang et al., 2015).

All experiments were conducted three times and had a control group (0 mg/L). One-way ANOVA was performed to assess differences between the treated groups and the control group. We used SPSS 21.0, Graphpad prism 6.01, MATLAB and Excel 2010 to obtain the statistical charts. The value for each parameter is the average of all of the repeated experiments. The data were presented as the mean $\pm$ standard error of the mean (SEM). $p < 0.05$ were considered to indicate a significant difference. And the statistical significances were set to $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), $p < 0.0001$ (****). Fast Fourier transform, edge detection and Watson–William test are implemented on MATLAB.

3. Results

3.1. Glyphosate exposure caused developmental retardation and increased mortality rate in developing zebrafish

We chose a wide range of concentrations aimed at determining the toxicity of glyphosate. The dynamic light scattering (DLS)

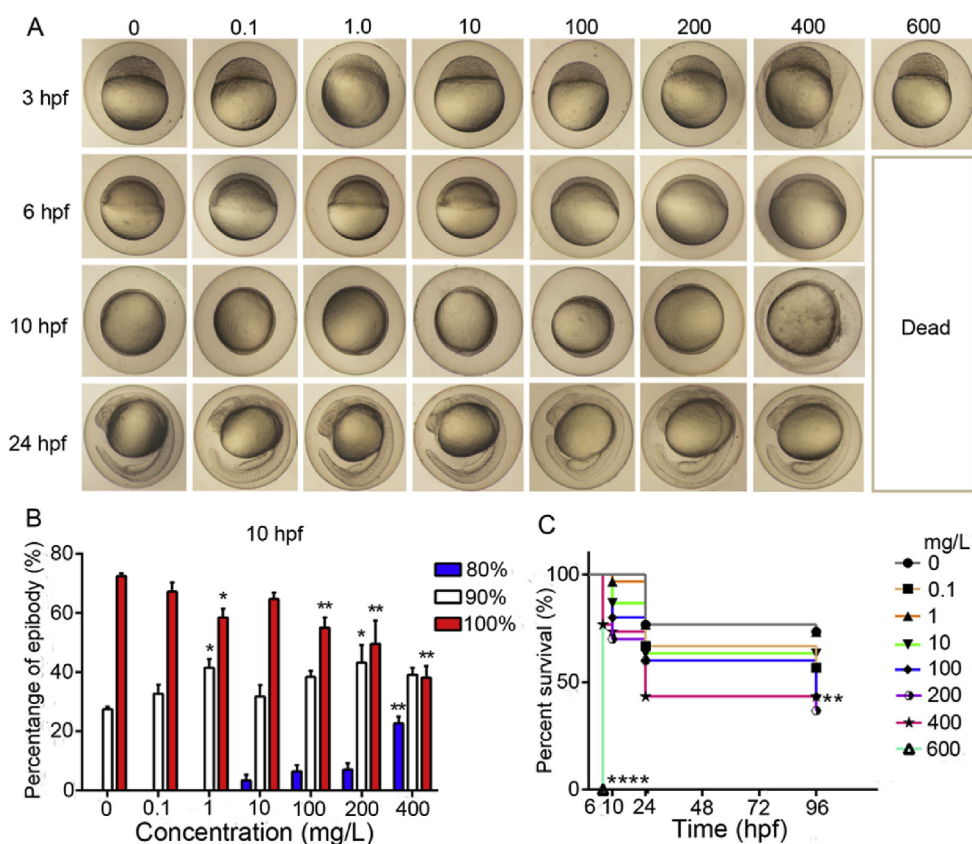


Fig. 1. Effects of glyphosate treatment on embryonic development. (A) Morphological characteristics of embryos exposed with glyphosate from 3 to 24 hpf. (B) The epiboly percentages of the embryos exposed to the glyphosate at 10 hpf. The error bar represents the standard error of the means (SEM). The statistical significances were versus the control group (C). The statistical significances were set to $p < 0.05$ (*) and $p < 0.01$ (**). (C) Survival evaluation: Kaplan–Meier plot representing the percent survival (%) that survived in 6 distinct time point: 6, 10, 24, 48, 72, 96 hpf. $n = 3$ replicate colonies per treatment and each replicate containing 40 embryos. *Indicate significant differences from control group (0 mg/L) survival. The statistical significances were set to $p < 0.05$ (*), $p < 0.01$ (**).

results showed that glyphosate didn't exist in the form of undissolved particles (Supplementary Fig. 1B). The zeta potentials of glyphosate dissolved in standard system water were electronegative (Supplementary Fig. 1C). The pH of the treatment solutions was about 7.0 when the concentration was <10 mg/L. But at higher concentrations (>100 mg/L), the solutions were acidic

(Supplementary Table 1). For example, the pH of solution was 3.9 ± 0.03 at 400 mg/L.

Glyphosate exposure was started at 0.25 hpf and end at 96 hpf. At the early 24 h, the development of embryos was recorded (Fig. 1A). No morphological alterations were found between the control group and the treated groups from 3 to 24 hpf at lower

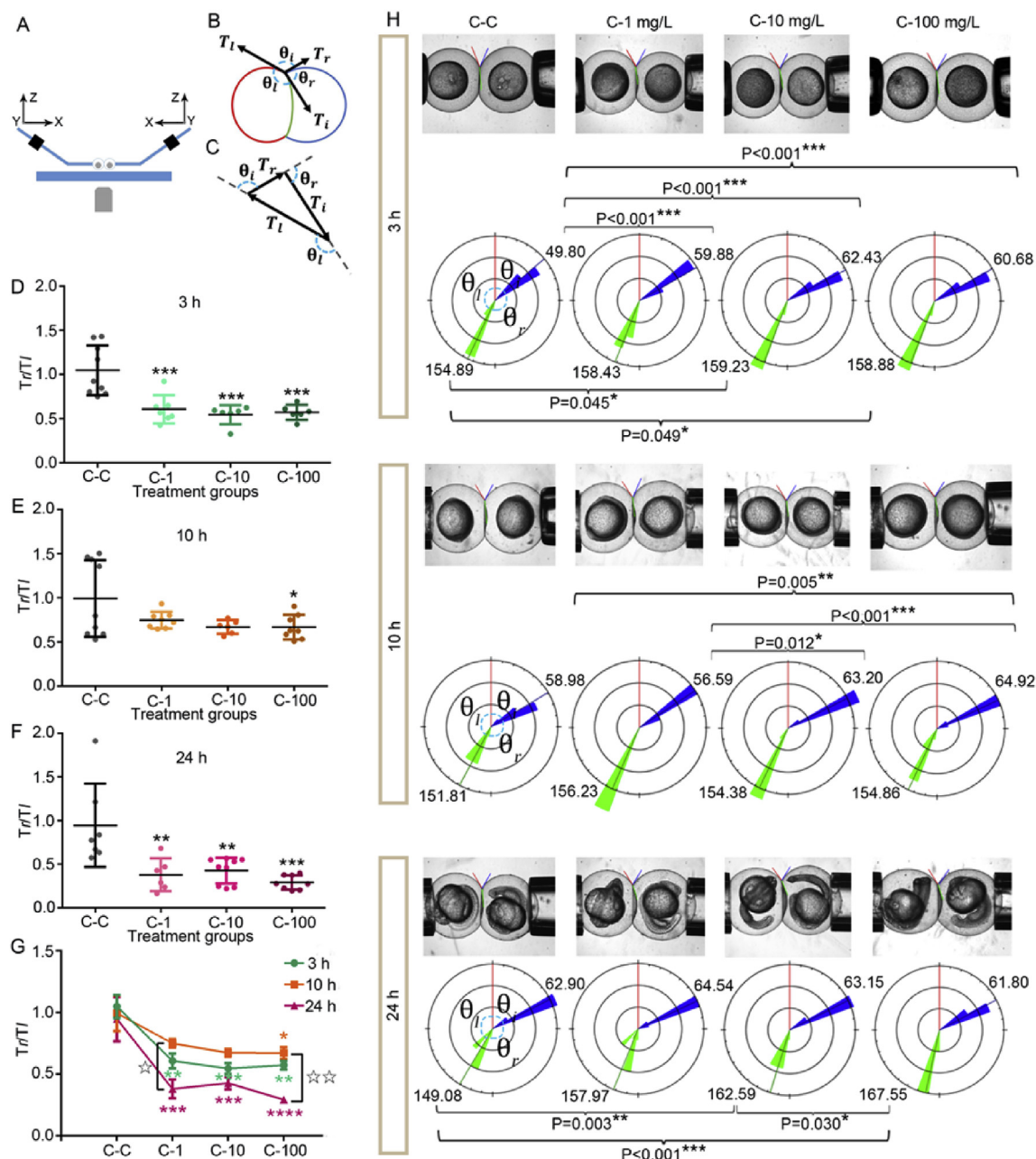


Fig. 2. Effects of glyphosate treatment on embryo surface tension. (A) Schematic diagram of experimental equipment. (B–C) Physical model of our test. There are three kinds of tensions exist on the edge of interface of two embryos, which are normal (left) chorion-water T_i , treated (right) chorion-water T_r , chorion-chorion T_c . The angles between the tangent of chorion-chorion interface and the tangent of left or right embryo-water interface at the edge of interface are θ_i and θ_r respectively. The angle between the tangents of two embryo-water surfaces at the edge of interface is θ_c . (D–F) The ratio of tension at two embryos' surfaces. "c" represents the control embryo. (G) The ratio of tension at two embryos' surfaces at different time points. $n \approx 6$. The error bar represents the standard error of the means (SEM). The statistical significances were set to $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), $p < 0.0001$ (****). The symbol pentacle (☆) represents statistical difference profile between different treatment time. (H) Micrographs of tested embryos at their final position (The embryo at left was the control, which was not exposed glyphosate, while the embryo at right was exposed to different glyphosate concentration. The red line represents tension at left chorion-water interface T_i . The blue line represents tension at right chorion-water interface T_r . The green line represents tension at chorion-chorion interface T_c .) and distribution of tangents of three chorions' surfaces. The red line is the tangents of chorion's surface of normal (left) embryos and set to 0° . The blue sectors are the range of chorion's surface of treated (right) embryos' surface. The green sectors are the range of interface of two chorions. The dark blue line and dark green line are their mean orientations respectively. The number is angle of the mean orientation. The radius of sector is the proportion of surfaces lie in the range corresponding to the sector and the radius of the three circles are 0.25, 0.5, 0.75. The three angles between each of two tangents are θ_i , θ_r , θ_c . *Indicate significant differences between tangents. The statistical significances were set to $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), $p < 0.0001$ (****). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

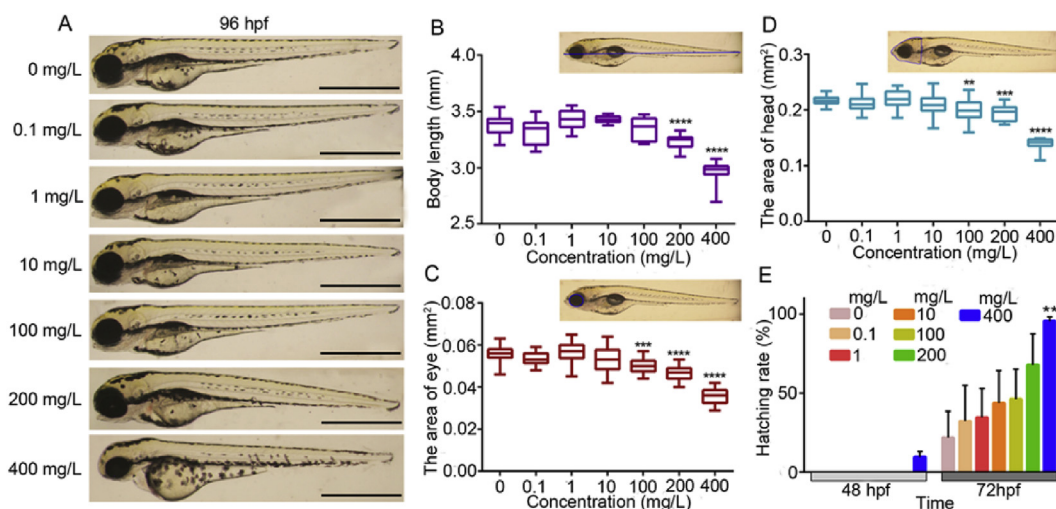


Fig. 3. Phenotypes of zebrafish larvae exposed to different concentrations of glyphosate and hatching rate of different time. (A) Morphological characteristics of control larvae and larvae exposed to different concentrations of glyphosate at 96 hpf. Scale bars = 1 mm. (B–D) Body length, eye area, head area of larvae. $n = 88, 68, 88, 76, 52, 44, 52$. (E) The hatching rate of larval treated with different concentrations at 48 and 72 hpf. $n = 92, 80, 92, 76, 72, 52$. The error bar represents the standard error of the means (SEM). The statistical significances were set to $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), $p < 0.0001$ (****).

glyphosate concentrations (0.1, 1, 10 mg/L). From 6 to 10 hpf, obvious delay in the epiboly process by higher glyphosate concentration (>10 mg/L) was detected. And significant embryonic abnormality (Fig. 1A) and mortality rates (Fig. 1C) were detected. Especially at 6 hpf, all embryos treated by 600 mg/L glyphosate were dead. Apparently, at 10 hpf, the epiboly process should be completed 100%. But the epiboly processes of some embryos at different concentrations were just up to 90%, even 80% (Fig. 1B). Delay in the epiboly process increased as glyphosate concentrations increased. So, the developmental retardation caused by glyphosate followed a concentration-dependent pattern at 10 hpf. With the development continuing, the morphological

characteristics of the zebrafish embryos which were exposed with higher glyphosate concentrations (>10 mg/L) at 24 hpf showed developmental delay compared with the control group (0 mg/L). As we can see, embryonic death occurred mainly in the previous 24 h.

3.2. Surface tension of chorion decreased because of glyphosate treatment

According to toxicity of glyphosate in developing zebrafish, we chose a series of concentrations to test the ratio of tension at a normal (control group) and a treated embryo's chorion (Fig. 2). When the test concentrations were less than 1 mg/L, the change of

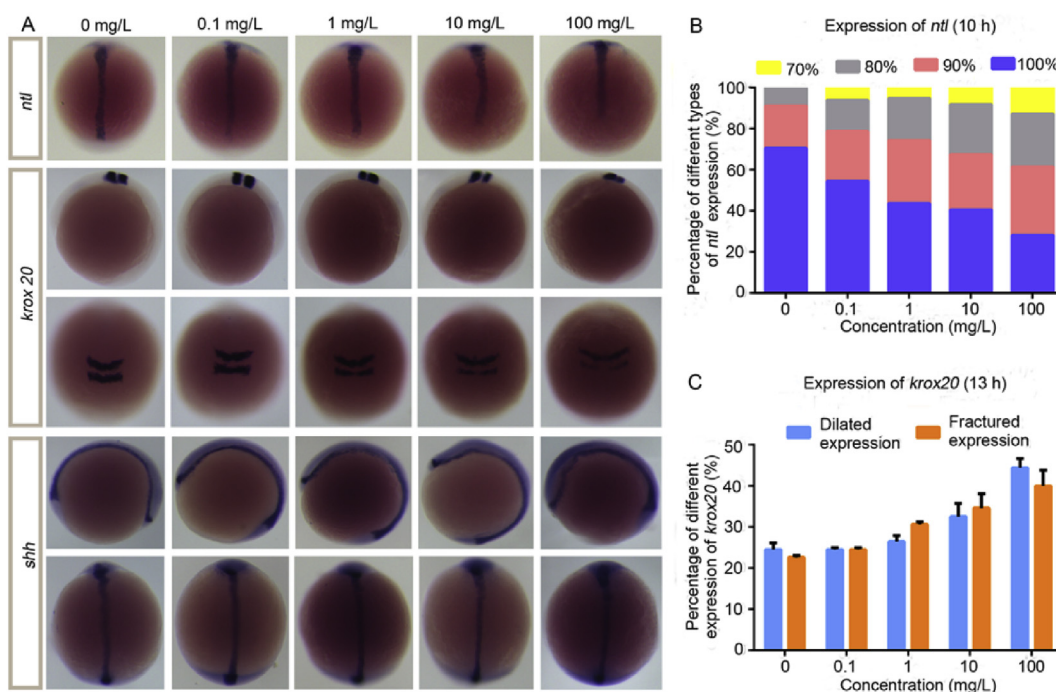


Fig. 4. Gene expression in embryos exposed to different concentrations of glyphosate. (A) Expression of *ntl* at 10 hpf, *krox20* at 13 hpf, and *shh* at 13 hpf in embryos exposed to 0 (C), 0.1, 1, 10, 100 mg/L glyphosate. (B) Statistical analyses of four types of *ntl* expression at 10 hpf in the controls and treated embryos. (C) Statistical analyses of two types of *krox20* expression at 13 hpf in the controls and treated embryos. The error bar represents the standard error of the means (SEM). $n = 40$ tested embryos.

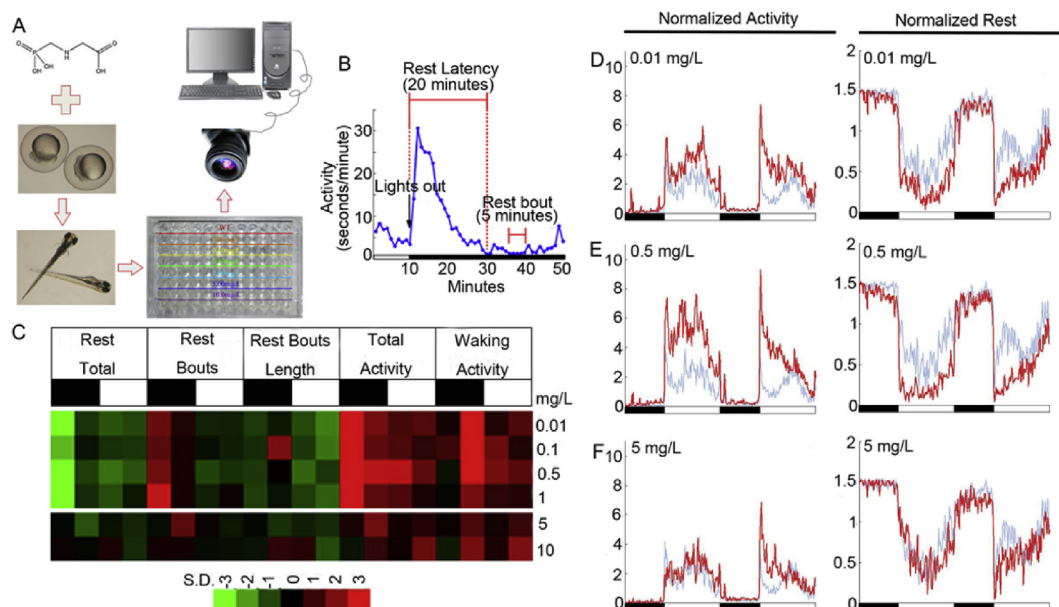


Fig. 5. Locomotor activity assay in larval zebrafish and behavioral similarities induced by different concentration of glyphosate. (A) The process for the diel activity behavioral assay. Glyphosate exposure was started at 0.25 hpf and ended at 48 hpf. Larva (4 dpf) was then pipetted into each well of a 96-well plate (~360 μ L) with different concentrations of glyphosate at 9:00 a.m. Zebrafish were observed beginning at 9:00 p.m. at 4 dpf. (B) We defined five parameters, including rest total, total activity, the number of rest bouts, rest bout length and waking activity from locomotor activity of a representative larva. (C) K-means clustering analysis. Each row represents a concentration of glyphosate, and each column represents a behavioral parameter. The black bars indicate the night measurements, and the white bars indicate the day measurements. These parameters were normalized as standard deviations from the control values. The red and green colors indicate that the values are higher and lower than the controls, respectively. (D–F) The waking activity and the rest total were averaged in 10-min intervals and then normalized to the control values. In the time series analysis, the red trace indicates the average of the treated group, and the blue trace indicates the average of the controls. The black and white bars represent the night and day measurements, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ration was not influenced by glyphosate. The data are not shown here. So we chose three (1, 10, 100 mg/L) treatment groups. The ratio of tension at two normal treated embryos' chorion approximately equal to 1 with approximately equal θ_l , θ_r . But the ratios of tension at a normal and a treated embryos' surfaces decreased with larger θ_l and larger difference between θ_l and θ_r as the concentration of glyphosate increased (Fig. 2B–H). At different treatment time, the ratio of tension at a normal embryo and a treated embryo was decreased significantly compared with two normal embryos, but there were no significant difference within different treatment concentrations (Fig. 2D–F). Furthermore, the decrease of the ratios became more significant with longer treatment periods, especially 24 hpf (Fig. 2G). In short treatment periods, the increase of θ_l and θ_r keeping constant caused the ratios decrease. But in long treatment periods, the increase of θ_l and decrease of θ_r caused the ratios decrease. It indicated that glyphosate caused decrease of the tension of embryo's chorion, but not at a concentration-dependent manner.

3.3. Glyphosate induced hatching rate increase and larvae abnormality

To further explore the effect of glyphosate on developing zebrafish, morphological assessment and analysis were conducted for larval development at 96 hpf and hatching rate was recorded at 48 hpf and 72 hpf (Fig. 3). From 100 to 400 mg/L, the glyphosate exposure resulted in shorter body lengths, smaller eyes and heads, especially in 400 mg/L (Fig. 3A), there were significant differences comparing to control group (Fig. 3B–D). Interestingly, exposure to higher concentrations of glyphosate increased hatching rate compared with the control group (Fig. 3E).

3.4. Glyphosate treatment changed gene expression patterns in embryo development

The morphological changes are closely related to gene expression patterns. We used Whole-mount *in situ* hybridization to test three key genes' expression, including *ntl*, *krox20* and *shh*. According to the test we conducted above, we chose a part of concentrations (0.1, 1, 10, 100 mg/L) to assess the effect of glyphosate on zebrafish embryos (Fig. 4). The *ntl* expression pattern in embryos treated with glyphosate was shorter than the control group as glyphosate concentrations increased (Fig. 4A and B). And the expression of *krox20* changed as glyphosate concentrations increased. Expression of *krox20* changed and the expression of *r5* was completely broken (Fig. 4A, C). However, the level of *shh* expression were spatially and temporally normal (Fig. 4A).

3.5. Larvae treated with glyphosate increased their locomotor activities in day time

Considering the low toxicity of glyphosate to zebrafish in general case, we tested the diel activity behavior of zebrafish larvae which were exposure to lower glyphosate concentrations previously in their early embryonic development. Embryos were reared in various concentrations of glyphosate (0.01, 0.1, 0.5, 1, 5, 10 mg/L) to 48 hpf, then moved to control solutions until 96 hpf, then the test was run from 96 hpf to 144 hpf. We run the behavior test at 9:00 PM at 96 hpf. According to Schier's method and previous study (Rihel et al., 2010; Wang et al., 2014), we utilized a high-throughput, quantifiable approach to record the larval locomotor activity (Fig. 5A). Thus we can obtain the locomotor activity data of larval diel activity behavior for 48 continuous hours. From these data, we extracted five parameters, including the rest total, the rest

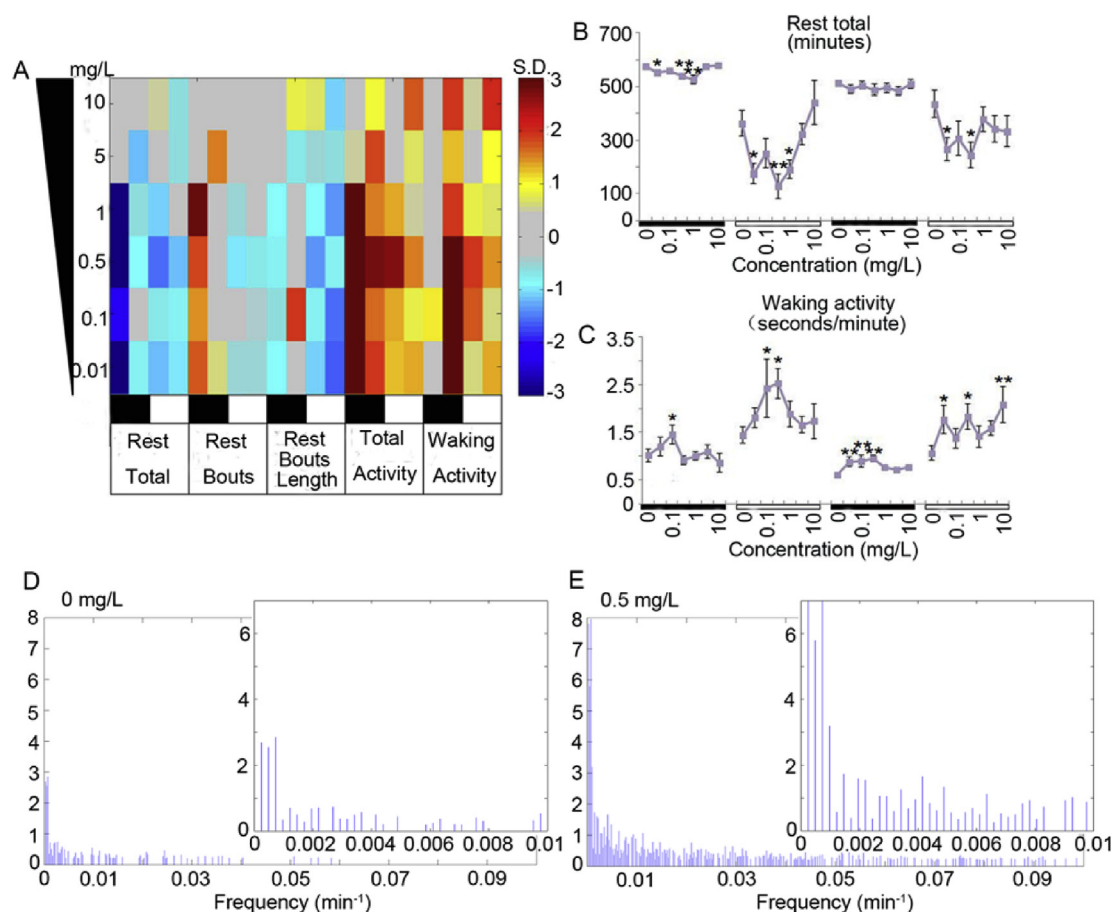


Fig. 6. Qualitative and quantitative analyses of behavioral changes induced by different concentrations of glyphosate. (A) In the fingerprint, five parameters were described. From left to right, the parameters were: rest total, number of rest bouts, rest bout length, total activity and waking activity. Each row represents a concentration of glyphosate, and each column represents a behavioral parameter. The measurements are normalized as standard deviations from the control values. The blue and red colors indicate that the values were lower and higher relative to the control values. The black bars indicate the night measurements, and the white bars indicate the day measurements. (B–C) In the line graphs for rest total and waking activity, each value indicates the average of ~36 larvae. The error bar represents the standard error of the means (SEM). The statistical significances were set to $p < 0.05$ (*) and $p < 0.01$ (**). The black and white bars indicate the night and day measurements, respectively. (D–E) Frequency domain graphs at concentrations of 0 and 0.5 mg/L. The graph in the top right corner is the detail of 0–0.01 min^{-1} . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

bouts, the rest bouts length, total activity and waking activity, to assess the diverse effects of different glyphosate concentrations (Fig. 5B). Based on the five normalized parameters, we conducted k-means clustering analyses according to shared behaviors. We chose $k = 2$ for the detailed analyses (Fig. 5C). As shown in the cluster-gram, 0.01–1 mg/L were co-clustered and the rest was another category. The clustering analysis demonstrated that the lower and the higher concentrations induced different behavioral phenotypes. Normalized activity and rest were shown in Fig. 5d–f and Supplementary Fig. 2. As we can see, the lower concentrations increased activity and decreased rest to varying degrees on two days (Fig. 5D and E). But the higher concentration mainly affected the two parameters slightly on the second day (Fig. 5F). Based on the time series, glyphosate seemed no to significantly affect the two parameters on night time.

In order to explore the more specific effects on larval behavior, five behavioral parameters were further quantitatively analyzed, including behavioral fingerprints (Fig. 6A), line graphs, histograms (Supplementary Fig. 4) and Fast Fourier transform. Compared with control group, the exposure groups decreased rest total time on days and the first night significantly. Waking activity increased inordinately in all treated groups on 48 continuous hours (Fig. 6B and C). Glyphosate only increased number of rest bouts on the first night

and decreased rest bouts length on the first night and day at 0.01–1 mg/L. Those two parameters changed irregularly and non-significantly (Supplementary Fig. 3). In short, glyphosate mainly increased larval activity on day time. We used Fast Fourier transform to investigate whether the zebrafish rhythms changed (Fig. 6D and E and Supplementary Fig. 5). All groups had the same main wave which had a 1200-min period and corresponded to the alternation of day and night. Compared with control group (Fig. 6D), the main wave of treated groups had greater amplitudes. Besides, we detected that many high-frequency but smaller-amplitudes waves which didn't exist in control group. It manifested that larvae exposed to glyphosate increased their locomotor activities, especially at 0.01–1 mg/L (Fig. 6E and Supplementary Fig. 5).

3.6. Glyphosate damaged motoneurons expression in larvae

Zebrafish have two types of spinal motoneurons, primary motoneurons (PMNs) and secondary motoneurons (SMNs) (Tallafuss and Eisen, 2008). Primary motoneurons are born early in development, around the end of gastrulation, while secondary motoneurons arise later in development and are more numerous than PMNs (Cheesman et al., 2004; Hutchinson and Eisen, 2006). Here, we focus on one of the PMNs, CaP, which has a ventrally projecting

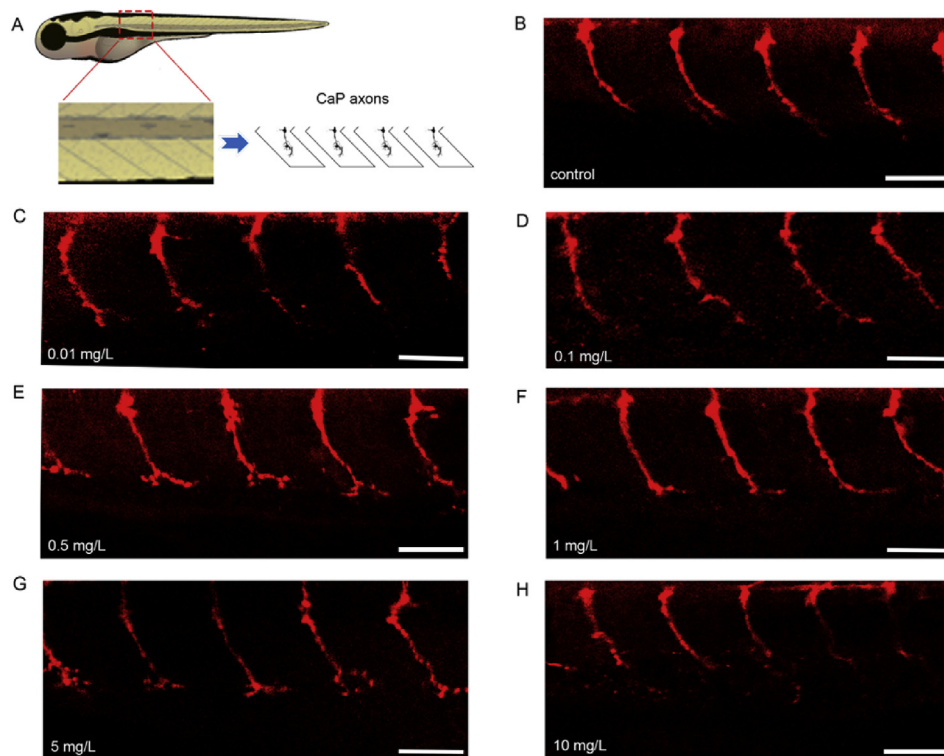


Fig. 7. The motoneurons expression at different concentrations of glyphosate. (A) Schematic showing expression of CaP axons. (B–H) 28 hpf larvae stained with znp1 Abs (red) have ventrally projecting CaP axons. Scale bars = 100 μ m. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

axon. From control group, we can see the motoneurons expressed in a regular pattern (Fig. 7B). With the increase of glyphosate concentrations, motoneurons became less and shorter, especially in 10 mg/L (Fig. 7H). There is no doubt that glyphosate damages motoneurons in 28 hpf.

4. Discussions

As a kind of herbicide used widely, evaluating biological effects of glyphosate is essential. Obviously, there is much research on this issue. But in our study, we combined developmental assay with biomechanics and zebrafish behavior profiling to assess the biological effects of glyphosate on zebrafish. We know that organophosphorous compounds, like glyphosate can induce organisms deformities (Yang et al., 2011; Faria et al., 2015). Here, development delay at concentrations higher than 100 mg/L from 3 hpf, and embryonic death mainly in the previous 24 h, showing glyphosate at higher concentrations could affect the early development of zebrafish in morphology. The high mortality of embryos was mainly attributed to the embryos' developmental retardation and abnormality. *No tail (ntl)* is one of important genes that are required for early notochord formation (Odenthal et al., 1996; Zou et al., 2009). We chose *ntl* as a marker for cell movement during gastrulation which was an important process in embryo development. *Krox20* (also known as *Egr2b*, early growth response 2b) is a zinc-finger transcription factor that is involved in vertebrate hindbrain segmentation, particularly in the formation and specification of rhombomeres (r) 3 and 5, as an important marker of zebrafish brain (Li et al., 2016b). *Shh* (sonic hedgehog) acts as a morphogen controlling multiple developmental processes (Schäfer et al., 2005). *Shh* expresses in midline structures such as prechordal mesoderm, the notochord and floor plate controls left-right asymmetry, neural survival, neuron identity and dorso-ventral

patterning of the neural tube during early vertebrate embryogenesis (Varjosalo and Taipale, 2008; Superina et al., 2014). In molecular level shown by *in situ* hybridization, the shorted expression of *ntl* demonstrated that glyphosate could reduce the structure of notochord, which is in accordance with the shorter body length in morphology. As shown in Fig. 4, the expression of *ntl* in notochord in 10 mg/L and 100 mg/L groups are shorter than that of the wild typed embryos. But the expression of *shh* of 10 and 100 mg/L groups are same to those of the wild typed embryos at 13hpf. *Shh* was expressed at notochord at 13 hpf, which suggested that the development of notochord was not affected by glyphosate of 10 and 100 mg/L. Thus the short expression of *ntl* only suggested that glyphosate of 10 and 100 mg/L could cause a delayed epiboly. The normal *shh* expression in notochord in embryos of 10 and 100 mg/L groups is in accordance with the body length in embryos of 10 mg/L and 100 mg/L, suggested that the glyphosate of 10 and 100 mg/L have no effect on the body length. The fractured expression of *krox20* demonstrated that glyphosate could change the gene expression patterns and interrupted the hindbrain segmentation, especially in the formation and specification of rhombomeres 5, as glyphosate concentration increased. High glyphosate concentrations (>100 mg/L) could cause larvae malformation. Likewise, there were no obvious morphological changes at lower concentrations. At the mean time, we found that treatment solutions were acidic at concentrations higher than 100 mg/L. Glyphosate caused an obvious decrease in pH. It was reported that the toxicity of glyphosate in water environment might come from its acidity when the glyphosate concentration was very high (Tsui and Chu, 2003). So the development toxicity might be related to the acidity of glyphosate solution at high concentrations. Since weak acid is as a chemical property of glyphosate, it's responsible for its toxicity to organisms.

Based on the development tests, we chose three glyphosate

concentrations to treat embryos, and test surface tension compared with a control group at 3, 10, 24 hpf. The results showed that surface tension of glyphosate treated groups decreased at a same time point in a non-concentration independent manner. As treated time extended, surface tension significantly declined, such as 24 hpf. However, no significant difference between 3 hpf and 10 hpf was found. In our opinion, surface tension's reduction resulted from that glyphosate destroys the structure of chorion, leading to brittle and easily broken of chorion. Interestingly, we found that higher hatching rate at 48 and 72 hpf than control embryos. It's speculated higher hatching in glyphosate treated embryos closely related with surface tension's reduction caused by glyphosate. Glyphosate exposure could decrease surface tension of chorion so that the hatching rates increased, which need further investigation.

We chose those lower glyphosate concentrations to simulate the real ecological environment. Behavior test showed that the glyphosate of 0.01 and 0.5 mg/L could increase the locomotive activities in day time. The concentration of 0.01 mg/L we used is close to environmental concentration of glyphosate, indicating the environmental glyphosate could change organism behaviors and increase spontaneous movement. Those changes might influence the response to danger and decrease the survival of organism under dangerous circumstance. With the concentration of glyphosate increased, the primary motoneurons were damaged severely which may be correlated with the changed locomotive activities alterations caused by glyphosate. Besides, it is reported that the increased hatching rates might be caused by the increased spontaneous movement at early stage (Jin et al., 2009). Thus, the damaged primary motoneurons by glyphosate might cause increased spontaneous movement, which further increased hatching rates. Hatching from chorion too early may expose organisms to contaminants, which is harmful to creature.

5. Conclusions

We hope that our research process can enrich the study of glyphosate. High concentrations of glyphosate can induce zebrafish development delay and embryonic death. The surface tension of chorion after glyphosate treated was decreased. The primary motoneurons, CaP axons, were damaged which may result in the increased locomotive activities alterations caused by glyphosate. The decreased surface tension of chorion and the increased locomotive activities may contribute to the hatching rates after glyphosate treatment. In summary, glyphosate have little harmful effect under lower concentrations on the early development of zebrafish embryos and larvae but may exert a potential effect on larval locomotor activities. Although the concentrations in natural aquatic environment is rather low, our study of evaluating toxicity of glyphosate at higher concentrations is necessary and probably plays a warning role in herbicides used in farming.

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The authors declare that there are no conflicts of interests.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.chemosphere.2017.04.094>.

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